RAHUL CHITHRA SHIVA

Fairfax, VA | +1 7033448820 | rchithra@umd.edu | https://www.linkedin.com/in/rahul-chithrashiva/

EDUCATION

University of Maryland, College Park — Expected Dec 2026
Master of Science in Applied Machine Learning
George Mason University (GMU), Fairfax, VA — May 2025
Bachelor of Science in Computer Science (dean's list) GPA - 3.57/4

TECHNICAL SKILLS

Languages: Python, Java (Data structures and algorithms), C (Computer Systems and Programming)**, C++,** Assembly language, SQL
Tools: AWS Cognito, AWS Lambda, Angular, Nest.js, PostgreSQL, Kivy, Buildozer, Flask, Dr. Java, GDB (debugging), GitHub, Apache Tomcat, Wireshark, FileZilla
Operating Systems: Windows, Macintosh, Linux, Onion OS
Applications: Visual Studio Code, Android Studio, MS Office Suite, Google Workspace, Zoom, Canva
Certifications: Programming using C and C++ (NIIT)

RELATED EXPERIENCE

Machine Learning Intern | Sasvaat Software Technol — Jun 2024 – Aug 2024
- Developed and integrated object detection functionality into mobile applications using YOLO, OpenCV, and Ultralytics, improving detection accuracy by 40% and reducing false positives by 15%.
- Successfully deployed object detection models using Kivy, Buildozer, and Android Studio, enhancing application performance and reducing deployment time. Utilized Ultralytics for advanced emotion detection.
- Collaborated with a team to integrate Chaquopy for using Python in Android development, enabling seamless integration of Python-based models and increasing development efficiency by 30%.

Full Stack Software Development Intern | Telecomatics Technologies — Jun 2024 – July 2024
- Implemented and optimized web application components using Angular, NestJS, and PostgreSQL in a collaborative team of 5 developers, resulting in a 30% increase in application performance and user engagement.
- Conducted detailed requirement analysis with the project management team for a new project, identifying key areas for improvement and contributing to a 20% reduction in development time.
- Delivered demos of developed components to clients, gathered feedback, and implemented changes, enhancing client satisfaction by 25% and ensuring alignment with client expectations.

Computer Science Teaching Assistant | George Mason University — Aug 2023 – Dec 2023
- Collaborated with a Graduate Teaching Assistant to guide 400+ Computer Science majors in CS 367 (Computer Systems and Programming), providing support in complex system processes such as CPU scheduling, forking, and piping
- Managed the Piazza discussion forum, addressing 100+ inquiries weekly with clear and concise explanations, which improved student understanding and engagement by 30%. Facilitated effective communication and problem-solving among students.
- Assisted students in creating their floating-point library and reverse engineering using assembly language, enhancing their practical programming skills and contributing to an increase in student project success rates

Math Learning Assistant | George Mason University — Aug 2022 – Dec 2024
- Assisted over 150 students in Analytic Geometry and Calculus lecture sessions by partnering with the course professor.
- Led 30+ weekly review sessions focused on exam preparation and concept mastery, increasing average student performance by up to 15%.
- Provided one-on-one assistance to 10–15 students weekly during office hours, addressing individual learning needs and boosting retention of core topics.

Clinic Software Intern | Child Kare Center — May 2022 – July 2023
- Developed and executed a comprehensive training program for clinic staff on billing, inventory, and pharmacy systems, resulting in increasing training efficiency.
- Collaborated with the IT team to identify, troubleshoot, and resolve software issues, enhancing overall system reliability and reducing downtime.
- Conducted regular audits of medication inventory using the new software system, leading to a 30% reduction in medication errors and significantly improving patient safety.

PROJECTS

Domain Name System(DNS) Client Server — MAR 2024
- Developed and implemented DNS Client protocol using Python and socket programming, Framed and sent DNS queries with DNS headers, questions, and other fields, facilitating domain name resolution to IP addresses.
- Monitored and verified DNS queries using Wireshark and analyzed, sent, and received packets to ensure accurate IP address resolution, enhancing reliability and efficiency in DNS communication.

Image Compression program — OCT 2023
- Applied expertise in data structures to enhance the algorithm's performance for image compression
- Engineered a highly optimized four-directional linked list, resulting in improved compression ratios and efficient storage utilization.

Reverse engineering using Assembly language — MAY 2023
- Implemented the AT&T syntax in the reverse engineering process, effectively navigating through intricate code structures and deciphering the bomb's logic to uncover the necessary passcodes.
- Utilized debugging tools, such as GDB, to analyze and reverse engineer the binary bomb, effectively identifying vulnerabilities and understanding the underlying code structure.

Advanced Shell Program — APR 2023
- Developed and implemented advanced functionality in a custom-built shell program, including forking processes, piping, background processes, and signal handling.
- Extensively worked on the internal workings of the shell program, ensuring seamless execution of commands and efficient handling of system resources

Floating Point Library — Mar 2023
- Developed and implemented a custom floating-point library using binary scientific notation in C, enabling accurate mathematical calculations on an embedded system without hardware floating-point support.
- Integrated the floating-point library concept into a custom Python-like environment, expanding the functionality and usability of the software for data analysis and manipulation tasks.
- Expanded the capabilities of the floating-point library by incorporating additional arithmetic support, enhancing the range of mathematical operations, and improving the accuracy of calculations within the embedded system.

Predictive Analytics for 15-Minute Ahead Blood Glucose Forecasting

- Built a **15-minute ahead blood glucose prediction model** achieving **~94% forecasting accuracy** with **8.15 mg/dL MAE**
- Engineered **lag-based features (60-minute historical window)** and trained **Random Forest & Ensemble models**, improving prediction accuracy by **~22% RMSE** over linear models
- Designed a **scalable ML pipeline** for preprocessing, training, and evaluation using **Python, Pandas, NumPy, and Scikit-learn**, enabling patient-level forecasting and visualization

F1 Lap Time Prediction

- Built an **end-to-end ML pipeline** to predict **Formula 1 lap times**, achieving **99.6% prediction accuracy** with **0.39s MAE** on same-event validation
- Engineered race and driver features and trained **Linear Regression and Random Forest models**, reducing prediction error by **66% vs baseline**
- Designed a robust evaluation framework using cross-event and holdout validation, ensuring model generalization across different tracks

BondBot

- Engineered ESP32 firmware in C using Arduino IDE to handle real-time touch input, Wi-Fi communication, and hardware signal processing across a 2-device paired embedded system.
- Architected and deployed a cloud messaging backend on DigitalOcean using Node.js, achieving **<0.5s end-to-end latency** for bidirectional presence and anxiety-regulation signal sync between paired devices.
- Designed and 3D modeled the full physical enclosure for both devices, integrating hardware layout constraints with embedded component placement to deliver a functional end-to-end prototype.

UnifyMed

- Developed a full-stack AI platform using Next.js and FastAPI that processes multilingual medical reports in any language, applying Gemini-powered OCR to extract and standardize clinical data at scale.
- Integrated RxNorm CUI mapping to normalize drug names across 100,000+ standardized codes, enabling consistent cross-border medical data interpretation regardless of source language or format.
- Designed and implemented the end-to-end API layer connecting frontend uploads to backend AI pipelines, delivering a working full-stack product within a 36-hour hackathon sprint.

Under Care

- Built a full-stack AI diagnostic platform supporting assessment across multiple injury and symptom categories, using GPT-4 for symptom reasoning and Torch-based computer vision for photo analysis.
- Engineered the backend inference pipeline in Python using Torch and Pillow for multi-category image classification, integrated with GPT-4 API to generate condition-specific diagnostic guidance.
- Shipped a responsive patient-facing frontend with Next.js and Tailwind CSS, taking the full product from zero to working demo across multiple use cases within a 24-hour hackathon sprint.